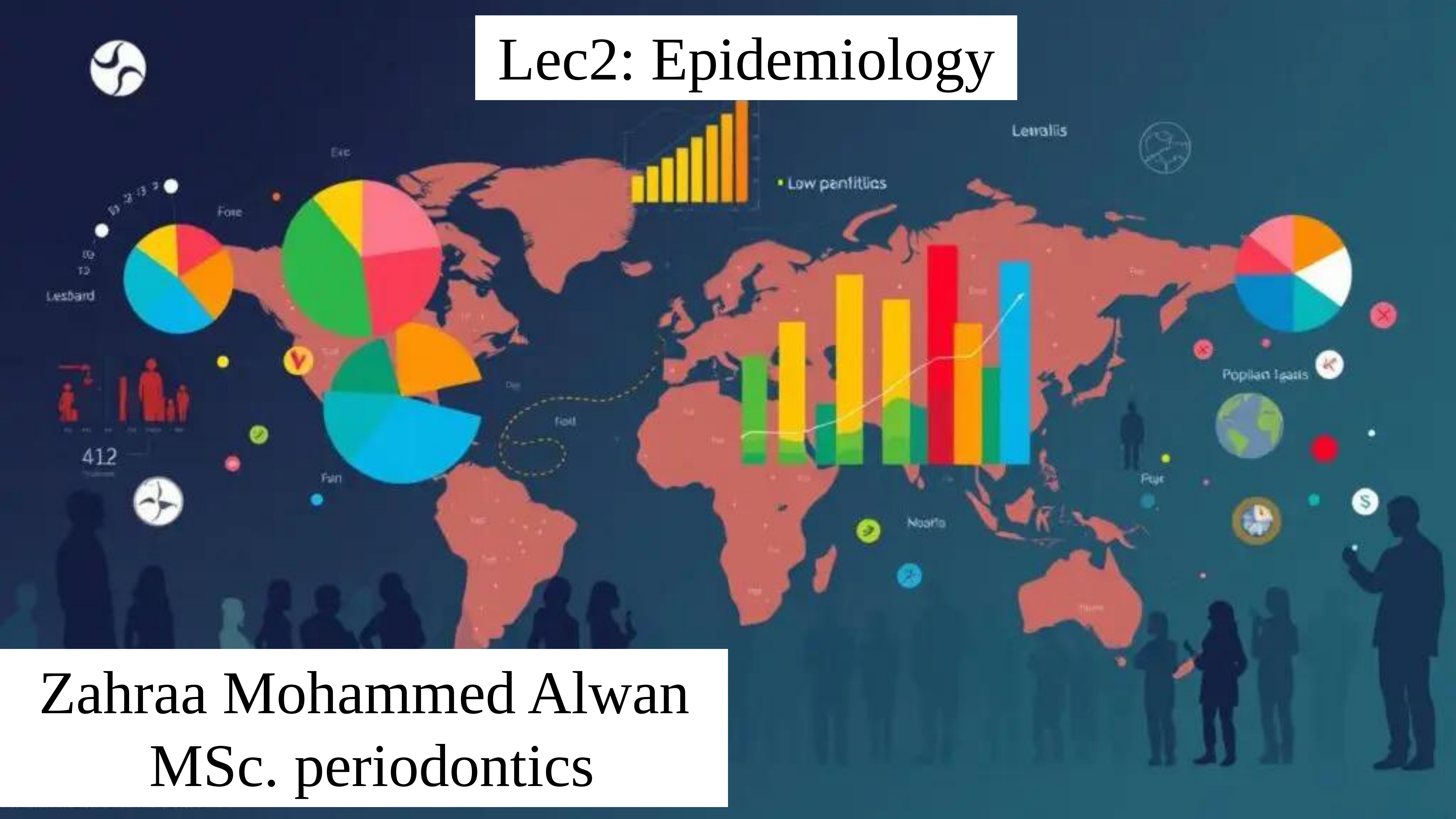
Lec2: Epidemiology



Zahraa Mohammed Alwan
MSc. periodontics

Epidemiology has been defined as the science concerned with the study of the distribution and determinants of a diseases and other health related events in specified populations, and the application of this study to control of health problems.

Objectives of epidemiology

1-To describe the distribution and magnitude of disease problems in human populations.

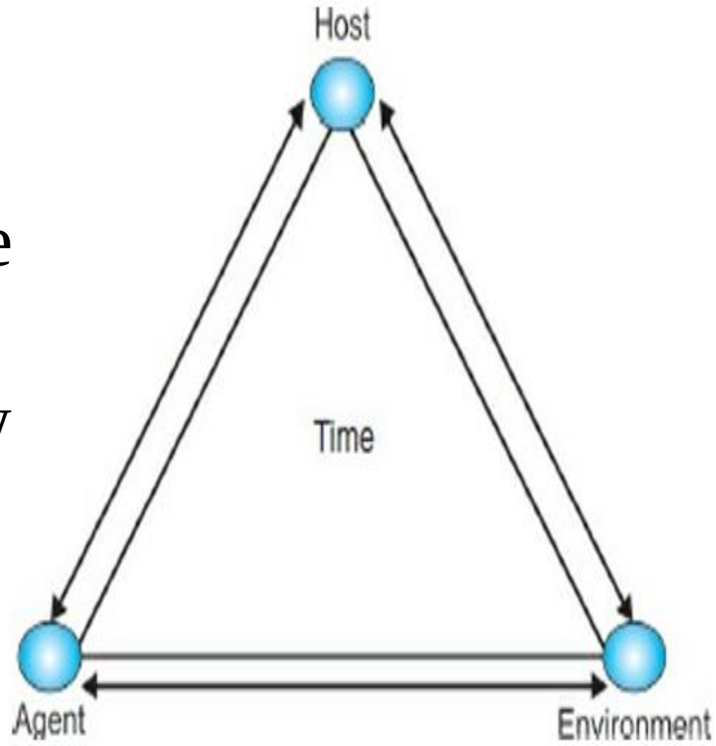
2-To identify etiological (or risk) factors in pathogenesis of a disease.

3-To provide data essential for planning, implementation and evaluation of services for the prevention ,control and treatment of disease.

Epidemiologic triad

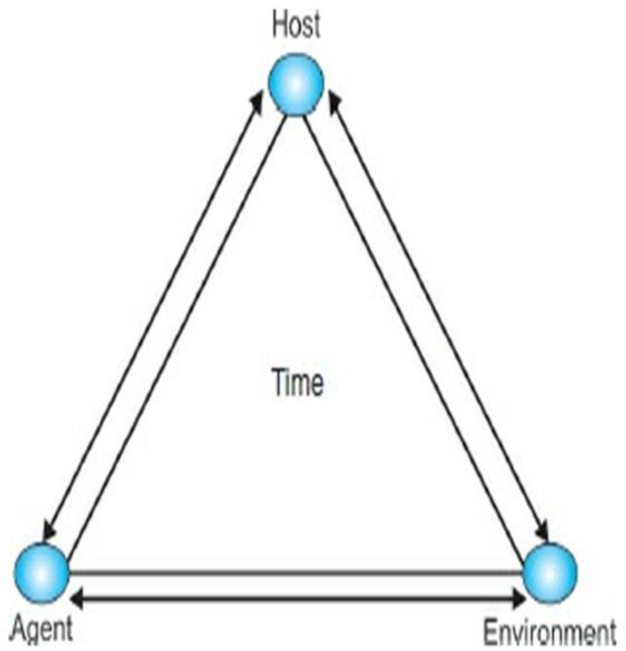
The epidemiologic triangle is a model that scientists have developed for studying health problems. It can help us understand infectious diseases and how they spread.

Agent, or microbe that causes the disease (the "what" of the triangle)
Host, or organism harboring the disease (the "who" of the triangle)
Environment, or those external factors that cause or allow disease transmission (the "where" of the triangle)

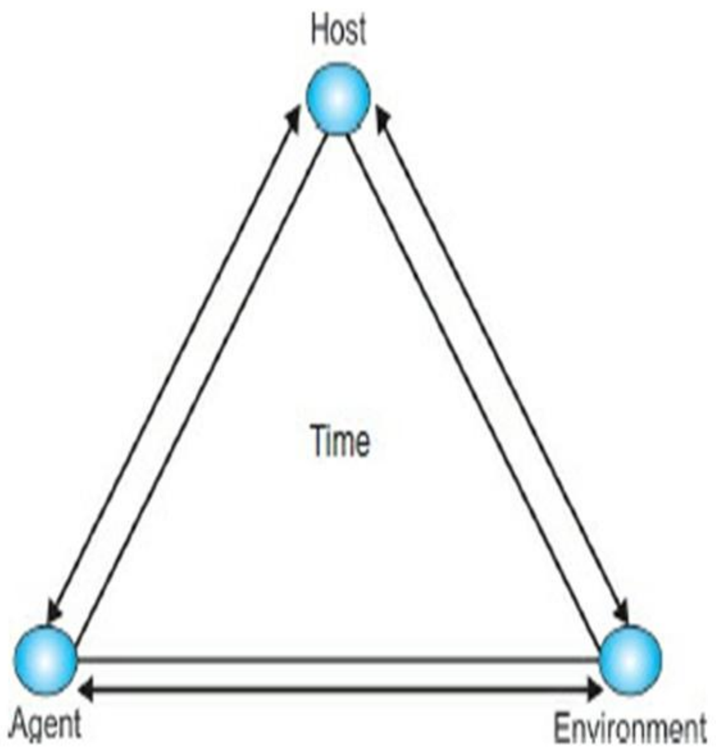


The agent—“what”. The agent is the cause of the disease. When studying the epidemiology of most infectious diseases, the agent is a microbe an organism too small to be seen with the naked eye. Disease-causing microbes are bacteria, virus, fungi, and protozoa (a type of parasite)

The host—“who”. Hosts are organisms, usually humans or animals, which are exposed to and harbor a disease. The host can be the organism that gets sick, as well as any animal carrier (including insects and worms) that may or may not get sick

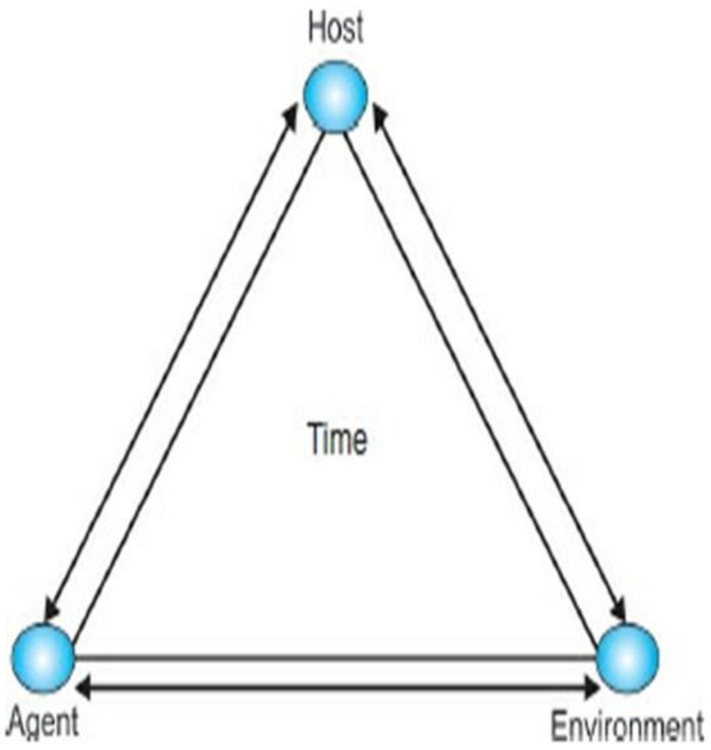


The environment—“where”. The environment is the favorable surroundings and conditions external to the host that cause or allow the disease to be transmitted. Still others, like E. coli, thrive in warm temperatures but are killed by high heat. Other environmentl factors include the season of the year (in the US, the peak of the flu season is between November and March, for example)



Time

In the center of the triangle is time. Most infectious diseases have an incubation period—the time between when the host is infected and when disease symptoms occur. Or, time may describe the duration of the illness or the amount of time a person can be sick before death or recovery occurs.



Strategy of epidemiology

1. Descriptive epidemiology :It includes description of the distribution of disease ,with comparison of its frequency in different populations and in different segments of the same population.

2.Formulation of hypothesis: tentative theories designed to explain the observed distribution of the disease in term of casual association of the most direct possible cause.

Strategy of epidemiology

3. Analytic epidemiology: observational studies designed specifically to examine the hypothesis developed as a result of descriptive study.

4. Experimental epidemiology: Experimental studies on human population to test in a stringent manner those hypothesis that resulted from the test of analytic studies.

Hypothesis: is a proposed explanation for a phenomenon. The hypothesis could be tested using the techniques of analytical epidemiology after which it may be accepted or rejected.

Measurement tools in epidemiology

The basic tools which are usually used to express condition or disease magnitude in epidemiology are:

1-Ratio –A ratio is obtained by dividing one quantity by another without implying any specific relationship between the numerator and the denominator. e.g. ratio of dentist to population is 1:10,000

Measurement tools in epidemiology

2-**Proportion** – is a type of ratio in which those who are included in the numerator must also be included in the denominator i.e., the numerator is a subset of the denominator. The magnitude of proportions is usually expressed as a percentage.

$$\frac{\text{Number of carious teeth at a certain time}}{\text{Total number of teeth present in the oral cavity at the same time}} \times 100$$

Measurement tools in epidemiology

3-Rate: It measures the occurrence of particular event (disease, death or extraction of teeth) in a population during a given period of time. The numerator is a part of denominator . Death rate is calculated as:

$$\text{Death rate} = \frac{\text{Number of death in one year}}{\text{Population}} \times 1000$$

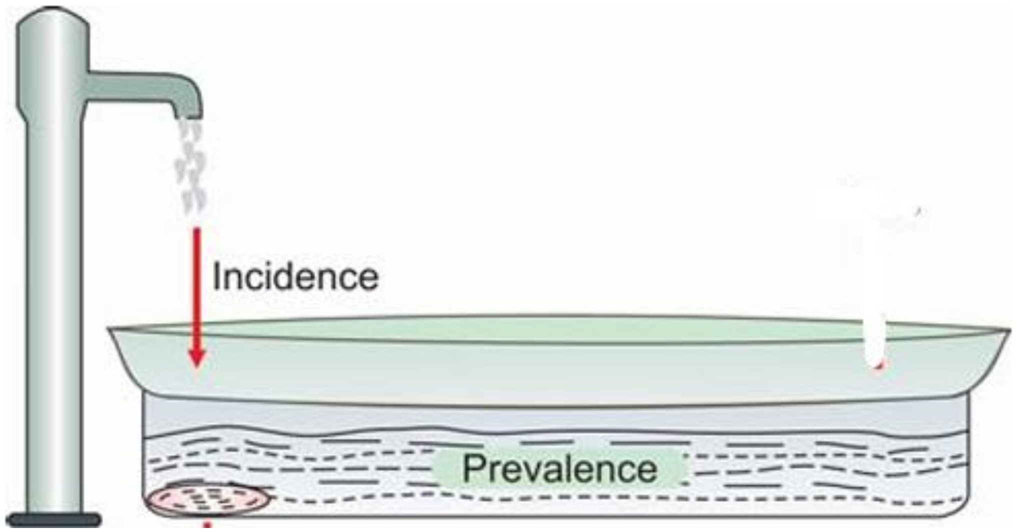
One of the most commonly used measurements in epidemiology is **Morbidity**, which is any departure from a state of physiological well-being.

Measurement of Morbidity

Morbidity is measured by two aspects: prevalence and incidence

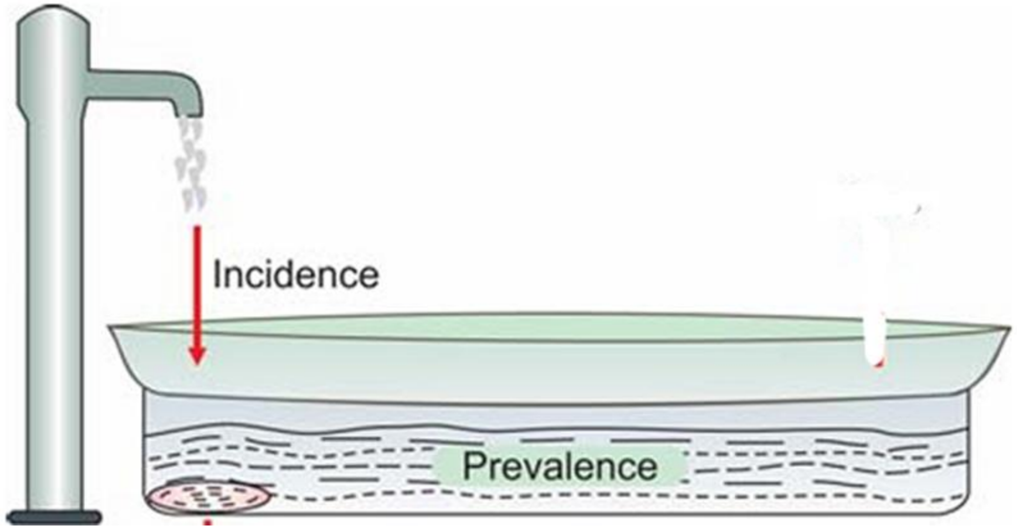
Incidence: measures the rate of appearance of new cases of a disease in a population divided by the population at risk during a given period of time (usually of a year).

$$\text{Incidence} = \frac{\text{Number of new cases}}{\text{Population at risk}} \times 1000$$



Prevalence: is defined as the number of all cases (old and new) in a defined population at a certain time point or over a period of time divided by population at risk.

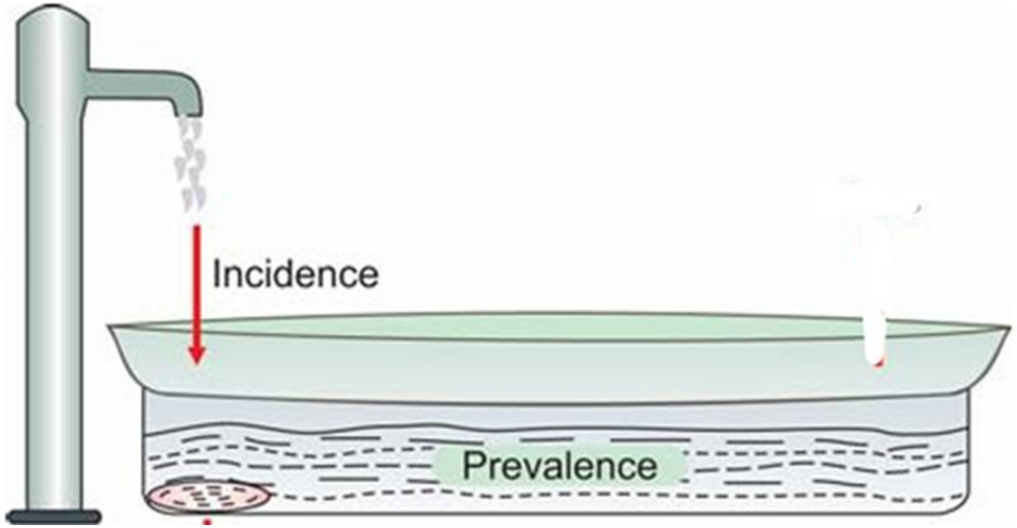
$$P = \frac{\text{number of all cases(old and new)}}{\text{Population at risk (the population Among whom the affected persons are observed)}} \times 100\text{(proportion)}$$



There are two types of prevalence:

(1) Point prevalence: measures all current cases of a disease at one point of time in relation to a defined population. The "point" in point prevalence, may consist of day, several days, or even a few weeks depending upon the time it takes to examine the population sample.

(2) Period prevalence: measures all current cases existing during a period of time (annual prevalence) expressed in relation to a defined population.




THANK YOU